

TRADEMARK AI SYSTEM

Complete Blueprint & Implementation Plan

Goal: Analyze trademark documents against TMEP guidelines → Output risk-categorized assessment reports

Source Document: TEAR, POUR, LIVE MORE.pdf | TMEP: tmept.uspto.gov (Nov 2025) | Date: Feb 2026

PART A — Your Two Questions Answered

Q1: Does the PDF contain trademarks? How to use it as training data?

YES — The PDF contains 5 layers of trademark data:

Layer	Data Source	Pages	Record Count	What It Contains
1	USPTO Federal Trademarks	p. 7 – 278	120 records	Mark name, serial no., registration no., status (Registered/Abandoned/Cancelled/Pending), International Classes, goods/services text, first-use dates, owner name & address, Section 2(d) refusal history, TTAB opposition history, assignment/ownership chain, Global Filings flag
2	State Trademarks	p. 279 – 310	25 records	State-level mark registrations (CO, VT, etc.), registration numbers, owner details, class coverage — used for common law prior rights analysis
3	Web Common Law	p. 311 – 318	5 records	Unregistered marks found via web search — brand names in commercial use without USPTO registration, establishing common law rights
4	Common Law Database	p. 319 – 330	5 records	Mintel Global New Products Database entries, trade directories, industry publications — business names using similar marks
5	Internet Domain Names	p. 331 – 443	61 records	Domain names incorporating TEAR, POUR, LIVE MORE — indicates real-world commercial usage and potential common law rights outside the USPTO
TOTAL	5 data sources	443 pages	216 records	Complete prior-art landscape for TEAR + POUR + LIVE MORE in Classes 05 and 32

The PDF also contains these critical metadata structures per record:

- Grouping System (Groups 1–5) — CompuMark's proprietary similarity scoring
- Analyst commentary by Johnny Sok — human-expert notes on what is risky and why
- Section 2(d) refusal cross-references — which marks were used to block others

- TTAB opposition outcomes — litigation history between marks
- Ownership chain / assignment history — who really controls the mark
- Correspondent attorney details — who to contact for licensing/disputes

How to use it as training data — 3 Approaches:

#	Approach	What You Extract	How AI Uses It
A	Structured JSON Records (Recommended)	Parse each citation into JSON: {mark_text, status, classes[], goods_services, serial_no, reg_no, owner, section_2d_history, group_number, first_use_date}	AI uses these as the 'TESS database substitute' — when you submit a new mark, it searches this JSON for conflicts, identical matches, phonetic matches, class overlaps
B	Fine-tuning Examples	Convert each citation + analyst notes into prompt-completion pairs: {input: 'Mark X in Class 5 conflicts with...', output: 'Risk: HIGH — TMEP §1207.01 — identical mark...'} }	Fine-tune a base LLM (GPT-4, Claude, Mistral) on these examples so it learns the pattern of trademark conflict assessment from real CompuMark analyst decisions
C	RAG Knowledge Base	Chunk the PDF into segments (one per citation), embed them into a vector database (Pinecone, Chroma, Weaviate), store with metadata tags (class, status, group)	When new mark submitted → embed it → retrieve top-K similar records from vector DB → feed retrieved records as context to LLM → LLM reasons about conflicts using real data

Best Choice for Your Goal: Use Approach C (RAG) for the knowledge base + Approach A (JSON) for structured conflict lookup. Approach B (fine-tuning) can be added later once you have 50+ PDFs.

Q2: How can you use the TEAR, POUR, LIVE MORE.pdf?

The PDF serves 4 distinct roles in your system:

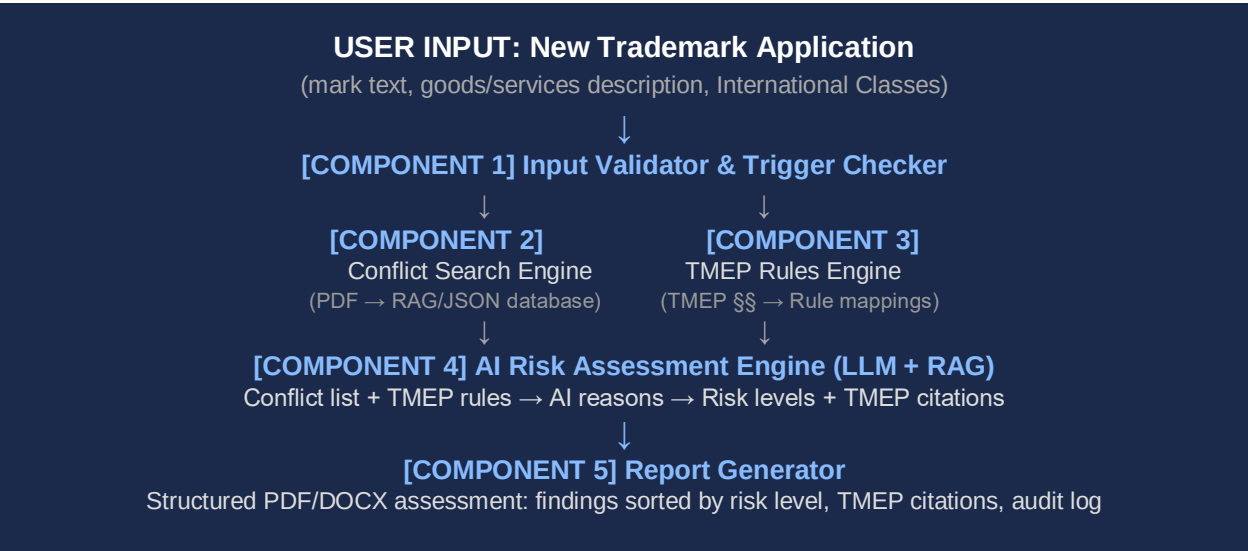
Role	Purpose	How It's Used	Output
1	Prior Art Database (TESS substitute)	120 USPTO records become your searchable conflict database. When a new mark is submitted, your system queries this data for identical, phonetic, spelling, and per-word matches	Conflict list with group classification (Group 1 = identical → Group 5 = weak similarity)
2	Risk Label Ground Truth	Group 1-2 = HIGH risk, Group 3 = MEDIUM-HIGH, Group 4 = MEDIUM, Group 5 = LOW. The CompuMark grouping IS your risk labeling system — already done by experts	Pre-labeled training examples: each citation already has a similarity group and analyst note
3	TMEP Anchor Points	Section 2(d) refusal data in the PDF maps directly to TMEP §1207.01 (likelihood of confusion). Each 2(d) citation is a real example of USPTO applying TMEP — perfect for grounding your AI's legal citations	Specific TMEP section → specific conflict type mapping

Role	Purpose	How It's Used	Output
4	System Validation / Test Case	Run your finished system against TEAR, POUR, LIVE MORE as the input mark. Compare your system's output to the CompuMark analyst's conclusions — this tells you if your system works correctly	Accuracy benchmark: does your system find the same 120 conflicts with the same risk levels?

PART B — Full System Architecture

The system has 5 core components that work together in a pipeline. The PDF feeds Component 2. The TMEP guidelines feed Component 3. Everything converges in Component 4 (the AI Engine) to produce the report in Component 5.

System Overview Diagram (Text)



Component 1 — Input Validator

What it does	Validates the incoming trademark application input and checks whether a search is legally required under TMEP
Required inputs	application_id, mark_text, mark_type (standard_character / design / sound), goods_services description, international_classes[], event_trigger
Valid triggers	first_review revival amendment_goods new_basis (anything else → system refuses with error)
TMEP reference	TMEP §704.02 — Mandatory search requirement before any office action
Output	Validated request object → passes to Component 2 & 3 in parallel

Component 2 — Conflict Search Engine (THE PDF ROLE)

Data source	TEAR, POUR, LIVE MORE.pdf → extracted and stored as structured records
Storage format	Two parallel stores: (1) JSON records for exact/structured queries, (2) Vector embeddings in a vector DB (e.g. ChromaDB) for semantic/phonetic search

Search types run	Exact match Phonetic match (Soundex/Metaphone) Spelling variants Per-word search Class overlap Goods/services semantic similarity
Source layers queried	Layer 1: USPTO (120 records) Layer 2: State marks (25) Layer 3: Web common law (5) Layer 4: Common law DB (5) Layer 5: Domain names (61)
Output	Ranked list of conflicting marks with: similarity score, group (1–5), source layer, all citation metadata → passes to Component 4

Component 3 — TMEP Rules Engine

Data source	TMEP Nov 2025 — https://tmep.uspto.gov/RDMS/TMEP/current — fetched, chunked, embedded into a separate vector DB
Key sections	§1207.01 (likelihood of confusion) §1207.01(a–d) (DuPont factors) §704.02 (mandatory search) §807 (mark description) §1402 (identification of goods) §1301 (service marks) §1202 (descriptiveness)
Rule mapping	Each conflict type maps to a specific TMEP section: identical mark + identical class = §1207.01(a)(i); phonetic similarity = §1207.01(b)(i); related goods = §1207.01(a)(v); famous mark = §1207.01(d)(i)
Output	Applicable TMEP sections retrieved as context for each detected conflict → passes to Component 4

Component 4 — AI Risk Assessment Engine

AI Model	Claude claude-sonnet-4-6 or GPT-4o via API — with structured output mode (JSON schema enforced)
Input to AI	(1) The new mark's details, (2) ranked conflict list from Component 2, (3) applicable TMEP sections from Component 3, (4) system prompt defining the 5 risk levels
Risk scale	HIGH MEDIUM-HIGH MEDIUM MEDIUM-LOW LOW — each level has defined criteria (see Part D)
AI tasks	For each conflict: assign risk level + cite specific TMEP section + explain reasoning + suggest mitigation action
Output	Structured JSON: [{finding_id, conflict_mark, risk_level, tmep_citations[], reasoning, mitigation}] → passes to Component 5

Component 5 — Report Generator

Input	Structured JSON findings from Component 4
Output formats	PDF report DOCX report JSON API response Audit log with unique ID, timestamps, all queries run
Report sections	Executive Summary Risk Heat Map Findings Table (sorted by risk) Detailed Citation Pages TMEP Reference Index Search Strategy Log Audit Trail
Cleared status	If no HIGH/MEDIUM-HIGH findings → mark <code>CLEARED_FOR_1207</code> (ready for substantive examination). Otherwise → <code>OFFICE_ACTION_LIKELY</code>

PART C — Extracting the PDF into Training Data

Step 1: Extract & Clean the PDF Text

- Tool: pdfplumber (Python) — already confirmed working on your PDF
- Process: Extract page by page, detect section boundaries by page header patterns
- Clean: Remove running headers ('TEAR, POUR, LIVE MORE'), page numbers, 'Search:411682111' footers
- Result: Clean text corpus, ~443 pages, organized by section

Step 2: Parse Each Citation into Structured JSON

Each of the 120 USPTO citations follows a consistent template. Parse it into this schema:

JSON Field	Data Type	Example / Source in PDF
<code>citation_id</code>	string	US-1, US-2 ... US-120
<code>mark_text</code>	string	"LIVE MORE"
<code>mark_type</code>	string	"standard_character" "design" "stylized"
<code>group</code>	integer	3 (from Group: Three)
<code>status</code>	string	"REGISTERED" "ABANDONED" "CANCELLED" "PENDING" "RENEWED"
<code>uspto_status</code>	string	"REGISTERED AND RENEWED"
<code>status_date</code>	date	"2020-10-06"
<code>international_classes</code>	integer[]	[5] or [29, 32]
<code>goods_services</code>	string	"Vitamins; dietary supplements; nutritional supplements..."
<code>first_use_date</code>	date	"2019-03-01"
<code>first_use_commerce</code>	date	"2019-03-01"
<code>owner_name</code>	string	"LIVEMORESUPERFOODS, LLC"
<code>owner_type</code>	string	"CALIFORNIA LIMITED LIABILITY COMPANY"
<code>owner_address</code>	object	{street, city, state, zip, country}
<code>serial_number</code>	string	"88-281,376"
<code>registration_number</code>	string	"6,170,206"
<code>filed_date</code>	date	"2019-01-29"
<code>registered_date</code>	date	"2020-10-06"
<code>section_2d_issued</code>	boolean	true
<code>section_2d_targets</code>	object[]	[{mark: 'SHOPLIVERMORE', status: 'ABANDONED', serial: '90754803'}]
<code>section_2d_cited_in</code>	boolean	false

JSON Field	Data Type	Example / Source in PDF
ttab_history	object[]	[{opposition_no, outcome, plaintiff, defendant, date}]
global_filings	integer	1
design_phrase	string	"THE MARK CONSISTS OF A CIRCLE ABOVE LIVE AND MORE..."
disclaimers	string[]	["NATURALS"] or []
assignment_history	object[]	[{assignee, assignor, date, brief, reel_frame}]
source_layer	string	"USPTO" "STATE" "WEB_COMMON_LAW" "COMMON_LAW_DB" "DOMAIN"
analyst_notes	string	CompuMark analyst comments if present

Step 3: Label Each Record with Risk Tier

Use the CompuMark group number + status + class overlap to auto-assign initial risk labels:

Group	Status Filter	Risk Label	Rationale
1 or 2	Any active	HIGH	Identical or near-identical mark — USPTO will almost certainly refuse
3	Registered + same class	MEDIUM-HIGH	Strong similarity to registered mark in identical class — high confusion risk per DuPont factors
3	Registered + related class	MEDIUM	Strong similarity but goods/services less directly overlapping — still warrants office action
4	Registered + any class	MEDIUM-LOW	Formative variant — possible confusion but DuPont factors may not fully support refusal
5 or Inactive	Abandoned/Cancelled	LOW	Weak similarity or inactive marks — monitor only, no immediate barrier

Step 4: Build the Vector Knowledge Base

1. Chunk the PDF — one chunk per citation record (avg 1–2 pages each)
2. Add metadata to each chunk: {citation_id, group, status, classes, mark_text, source_layer, risk_label}
3. Embed each chunk using an embedding model (OpenAI text-embedding-3-large or sentence-transformers)
4. Store in a vector database: ChromaDB (free, local) or Pinecone (cloud) or Weaviate
5. Index by both semantic content AND structured filters (class number, status, group)
6. Also create a separate embedding index for the TMEP Nov 2025 content from tmeptm.uspto.gov

Step 5: Create Fine-tuning Examples (Optional — Phase 2)

Convert each CompuMark citation into a prompt-completion pair using this template:

PROMPT: "A new trademark application for [MARK X] in Classes [A, B] covering [goods/services description] is being reviewed. Citation US-1 (LIVE MORE, Registered, Class 5, Owner: LiveMore Superfoods) was found in the prior art search. Does this citation pose a conflict?"

COMPLETION: "Risk Level: HIGH. TMEP Citation: §1207.01(a)(i). Reasoning: The cited mark LIVE MORE (Reg. #6,170,206) is identical in text to the applied-for mark. It covers Class 5 vitamins and dietary supplements – identical goods. It is actively registered and has demonstrated enforcement via Section 2(d) refusals. Under the first DuPont factor, identical marks on identical goods create a strong presumption of likelihood of confusion. Mitigation: Applicant should consider a different mark or a consent agreement with LiveMore Superfoods, LLC."

PART D — Risk Categories & TMEP Mapping

Risk Level	Trigger Condition	Primary TMEP §	What It Means	AI Output Action
HIGH	Identical mark + same/related class, active registration or pending	§1207.01(a)(i); §1207.01(b)(i)	USPTO will almost certainly issue a Section 2(d) refusal citing the blocking mark. Not registrable as-is.	Flag as LIKELY_REFUSAL. Suggest abandonment, redesign, or consent agreement.
MEDIUM-HIGH	Similar mark (Group 3) + same class; or identical mark + related class	§1207.01(a)(v); §1207.01(b)(ii–iii)	High probability of 2(d) refusal. Examiner will weigh DuPont factors — likely to find confusion.	Flag as PROBABLE_REFUSAL. Recommend legal counsel review DuPont analysis.
MEDIUM	Formative similarity (Group 4) + overlapping class; or similar mark + less-related goods	§1207.01(a)(i–vi); §1207.01(c)	Possible office action. Examiner may issue a 2(d) refusal — outcome depends on specific DuPont factor analysis.	Flag as POSSIBLE_REFUSAL. Conduct full DuPont analysis. Consider goods/services amendment.
MEDIUM-LOW	State mark / common law / domain in same field; or weak formative similarity	§1207.01(d); §904.03(f)	Low USPTO refusal risk but potential common law infringement claim from prior user. Not an absolute bar.	Monitor. Note for attorney review. Check actual marketplace use.
LOW	Inactive marks / Group 5 / different goods / domain names only	§1207.01(d)(iv); §904.03	Minimal current risk. Abandoned/cancelled marks are not bars to registration. Record for completeness.	Note in report. No action required. Mark cleared for §1207 analysis.

Key TMEP Sections Your System Must Reference

TMEP Section	Topic	When Your System Cites It
§704.02	Mandatory search before office action	Always — every search is legally required under this section
§1207.01	Likelihood of confusion — main section	Any mark similarity finding, any class overlap
§1207.01(a)(i)	Similarity of marks in appearance, sound, meaning	Group 1–3 citation found
§1207.01(a)(v)	Relatedness of goods/services	Different class but related goods found

TMEP Section	Topic	When Your System Cites It
§1207.01(b)(i)	Phonetic similarity	Soundex match found (e.g. TEAR vs TIER)
§1207.01(b)(ii)	Connotation / commercial impression	Same meaning in different words found
§1207.01(c)	Strength of cited mark	Famous or heavily-used mark found (amplifies risk)
§1207.01(d)(i)	Famous marks get broader protection	Mark with global filings or TTAB wins found
§1207.01(d)(iv)	Abandoned marks not bars to registration	Abandoned/cancelled citation being noted
§1202.02	Merely descriptive marks	Mark contains descriptive terms (LIVE, MORE, POUR)
§1402.01	Identification of goods/services	Goods description amendment recommended
§807.07(a)	Mark description requirements	Design mark or stylized mark found

PART E — Implementation Roadmap

Phase	Name	Duration	Key Deliverable	Tasks
1	Data Pipeline	Week 1–2	120 citations in JSON + vector DB	Extract PDF → clean text → parse JSON → label risk → embed → store in ChromaDB
2	TMEP Engine	Week 2–3	TMEP knowledge base + rule mapping table	Scrape/fetch TMEP Nov 2025 → chunk by section → embed → store → build §→conflict-type mapping JSON
3	Search Engine	Week 3–4	Working conflict search (all 4 query types)	Build exact match, phonetic (Soundex), spelling variant, per-word, semantic (cosine similarity) search functions against the vector DB
4	AI Engine	Week 4–5	Working risk assessor with TMEP citations	Prompt engineering → structured output schema → Claude/GPT-4 integration → risk level assignment → TMEP citation → mitigation suggestions
5	Report & Validation	Week 5–6	Full report output + accuracy benchmark	Report generator (PDF/DOCX) → run TEAR POUR LIVE MORE as test case → compare to CompuMark output → measure accuracy → iterate

Technology Stack

Layer	Recommended Tool	Purpose
PDF Extraction	pdfplumber (Python)	Already confirmed working on your PDF — extract text page by page
Data Storage (structured)	PostgreSQL or SQLite	Store the 120 JSON citation records for exact/filtered queries
Data Storage (semantic)	ChromaDB (free, local)	Vector embeddings for semantic similarity search
Embedding Model	OpenAI text-embedding-3-large	Convert mark text and goods/services into vectors
AI Engine	Claude claude-sonnet-4-6 via Anthropic API	Risk assessment, TMEP citation, reasoning — structured JSON output
Phonetic Matching	fuzzy / jellyfish (Python)	Soundex/Metaphone for POUR vs PORE, TEAR vs TIER
Report Generation	reportlab + python-docx	Generate PDF and DOCX assessment reports
API Layer	FastAPI (Python)	REST endpoints: POST /analyze → returns JSON assessment

Layer	Recommended Tool	Purpose
Orchestration	LangChain or plain Python	Connect all components in the pipeline
Validation / Testing	pytest	Test each search type against known CompuMark results

PART F — What the Final Report Looks Like

When a user submits a new mark (e.g. TEAR, POUR, LIVE MORE), your system generates a report with this structure:

#	Finding	Risk Level	TMEP Citation	AI Reasoning + Action
1	LIVE MORE (Reg. #6,170,206, Cl. 5) — LiveMore Superfoods	HIGH	§1207.01(a)(i)	Identical mark, identical class, active registration with prior enforcement. Near-certain 2(d) refusal. → Seek consent agreement or redesign mark.
2	LIVE MORE (Reg. #6,119,781, Cl. 29, 32) — LiveMore Superfoods	HIGH	§1207.01(a)(v)	Identical mark, Class 32 covers energy drinks — same goods as applicant. Enforced against LIQUID GOLD ENERGY DRINK. → Refusal expected.
3	LIVE MORE HEALTHY DAYS (Reg. #6,608,024, Cl. 5) — Cargill	MED-HIGH	§1207.01(b)(ii)	LIVE MORE is dominant element of cited mark. Cargill is large corporate registrant. Class 5 overlap. → Likely 2(d) refusal citing this as secondary block.
4	MORLIFE (Reg. #5,635,530, Cl. 29, 30, 32) — Las Ninas Trust	MEDIUM	§1207.01(a)(i)	MORLIFE is phonetic/anagram variant of LIVE MORE. Class 32 overlap on beverages. Has been successfully enforced via 2(d) against MORELIFE. → Possible secondary refusal.
5	Domain: DREAMBIGLIVEMORE.COM	MED-LOW	§904.03(f)	Domain with LIVE MORE formative indicates commercial use. No USPTO bar but potential common law issue. → Monitor.

This plan was generated from analysis of TEAR, POUR, LIVE MORE.pdf (CompuMark File 411682111) and TMEP Nov 2025. It covers Q1 (PDF structure + training data approaches), Q2 (how to use the PDF), and the complete AI system design. Ready to implement any component — just say which Phase to start with.